Public Transportation and Urban Planning

Instructions

In this worksheet, you'll practice writing **tidyverse-style R code** for data wrangling tasks as well as practicing data viz skills using `ggplot`

Don't stress yourself out over perfect syntax – the goal is to practice what you've learned about data wrangling and visualization so far.

Question 1

```
# A tibble: 1 x 75
  NTD.ID Agency        Organization.Type Mode  Type.of.Service Rail..True.False.
   <dbl> <chr>         <chr>             <chr> <chr>           <lgl>
1  90044 City of Arca~ County or Local ~ DR    PT              FALSE
# i 69 more variables: Primary.UZA.UACE.Code <dbl>, Primary.UZA.Name <chr>,
#   Primary.UZA.Sq.Miles <dbl>, Primary.UZA.Population <dbl>,
#   Service.Area.Sq.Miles <dbl>, Service.Area.Population <dbl>, Year <dbl>,
#   Month <chr>, Month_Year..timestamp. <dbl>, VOMS <dbl>,
#   Vehicle.Revenue.Miles <dbl>, Vehicle.Revenue.Hours <dbl>,
#   Unlinked.Passenger.Trips <dbl>, Collisions.with.Motor.Vehicle <dbl>,
#   Collisions.with.Person <dbl>, Collisions.with.Fixed.Object <dbl>, ...
```

Discuss the following questions with a partner our group:

- *What are the variables?*
- *What kinds of questions can we ask?*

Question 2

What does ridership over time look like for 1 specific Transit Agency? Use tidyverse data verbs and ggplot to wrangle and visualize the `ca_transit` data

- Filter data for specific Agency -> "Pomona Valley Transit Authority"
- Create line plot of `tot_pass` over time

Question 3

Write **ggplot** code to answer the following questions by plotting the relationship between the two variables.

(a) Do higher-income areas have more or fewer transit stops?

Plot the variables `mdn_ncm` and `stop_count` on a scatter plot with a line of best fit. What will is mean if the relationship on the graph is **positive**? **Negative**?

(b) Do areas where more people don't own cars have better transit coverage?

Plot the variables `pct_w_c` and `stop_count` on a scatter plot with a line of best fit. What will this mean if the relationship on the graph is **positive**? **Negative**?